

Class starts in 5 minutes

Agenda

- ↳ Project that we will build ✓
- ↳ What are frameworks & why
- ↳ Why Spring -
- ↳ Dependency Injection } Core of Spring
- ↳ Inversion of Control }
- ↳ Create our first Spring Boot App.

Project

- ↳ Basic E-commerce Platform (Backend)

Features in Flipkart

- ↳ Search ✓
- ↳ Product Detail ✓
- ↳ Payment ✓
- ↳ Login / Sign up ✓ (auth) ⇒ login through "Flipkart"
- ↳ Wishlist X
- ↳ reviews X
- ↳ cart X
- ↳ recommendations X

Less on implementation, more on concepts.

Tools to be installed

- ↳ Postman
- ↳ IntelliJ IDEA ultimate
- ↳ github pack.

What are frameworks & why we need them?

Common things as some

- ↳ api / endpoints

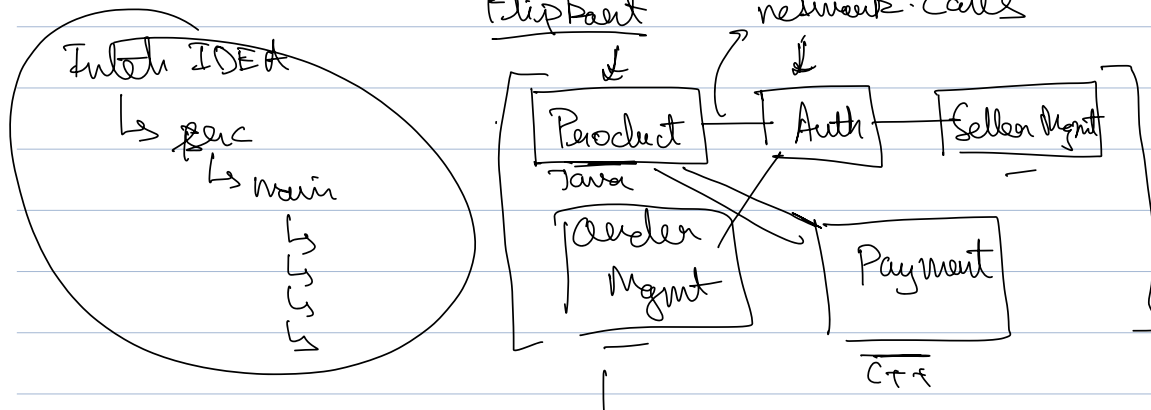
escaler.com / mentor-dashboard $\Rightarrow$ Sent the data

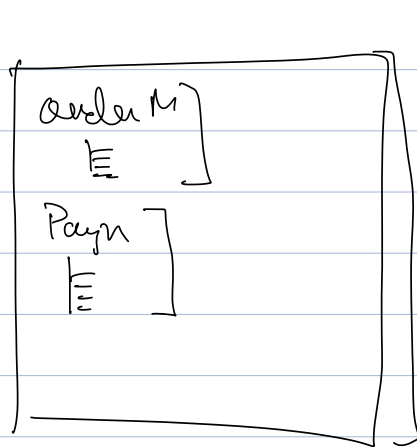
- ↳ interact with DB.

- ↳ cache (Redis)

- ↳ Messaging Queue (Kafka)

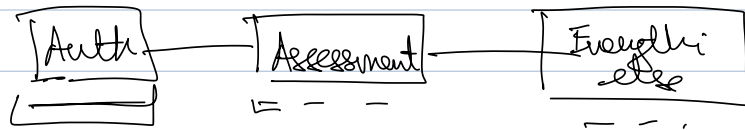
Microservice





↳ Diff services can have diff programming languages.

→ Monolithic architecture (single codebase)



Common things as sme

↳ api / endpoints

teacher-com / mentee-dashboard ⇒ Sent the data

↳ interact with DB.

↳ cache (Redis)

↳ Messaging Queue (Kafka)

↳ Microservices

I can write all the code for this but it will be very complicated ⇒ it won't be optimal use of my time.

Frameworks $\Rightarrow$ easy to use / efficient way to do common things.

$\hookrightarrow$ allow us to stand on the shoulder of giants
 $\hookrightarrow$ Don't reinvent the wheel.

B

Java: Spring, Dropwizard

Python: Flask, Django

C#: .net

Ruby: Ruby on rails $\Rightarrow$ Scalor

Our module $\Rightarrow$ Java + Spring

$\hookrightarrow$ Commonly used in the Industry

Why? $\Rightarrow$ 80% uses Java

15% Python

5% JS / C++ / C#

Tried & Tested tech

Every class $\Rightarrow$ Concept + implementation

$\Downarrow$

lang

independent

$\Downarrow$

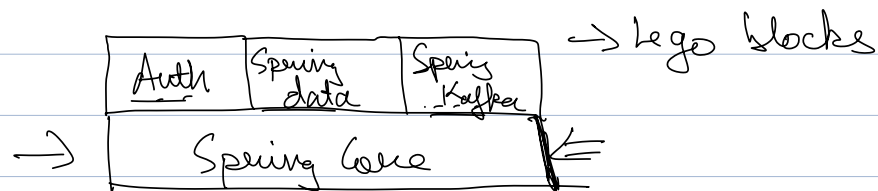
Java

Why Spring

↳ Very VVVV famous framework for building enterprise grade java app

↳ huge focus on scalability & security.

How? => Spring provides multiple projects/modules that make it easy for you to do diff set of things



Spring Core

↳ Dependency Injection

↳ App Context

Break for 7 minutes : 10:30 → 10:37

scaler.com/mentee-dashboard

Spring web]
Spring data
Spring security]
OAuth 2
Spring Cloud

① Providing the configuration for spring projects was a tiresome task.

```
<xml>
```

```
<name> MyDB </name>
```

```
<type> my sql </type>
```

```
...
```

```
</xml>
```

② Diff versions of diff modules/projects didn't work well with each other -

(Spring Security => secure apis Spring web	2.0.1	] 1.9.0
	3.0.2	
	3.0.0	

Spring Boot => Makes it very easy for me to work with spring -

How?

① Manages versions of diff modules very well

② Opinionated view.

→ "spring-data" ←

"spring-boot-starter-data" ← 3.2.2

→ "spring-boot-starter-web" 3.2.2

→ "spring-web"

Same version of diff "starter" projects work well with each other.

Timesome Configs

↳ Opinionated view

↳ spring boot sets some default configurations for us.

eg) ~~MySQL~~: Spring data

Data type ⇒ mysql

url ⇒ local host

port ⇒ 3306 ←

name ⇒ mydb

Only set the configs whose default value you don't agree with.

Good frameworks have an opinionated view but ~~are~~ make it easy to ~~for~~ change the options.

Spring Core

<pre>class LoginService { <u>Database d = new MySQL</u> <u>PostgreSQL</u> }</pre>	<pre>class SignupService { Database d = new MySQLDB(); PostgreSQL }</pre>	<pre>class UserDetails { Database d = new MySQLDB(); PostgreSQL }</pre>
---	---	---

Scenario => change from MySQL to PostgreSQL
↳ (problems)

Happening because each class is creating its own dependencies.

To solve this problem we can inject the dependency from outside:-

- ① constructor or,
- ② setter

```

class LoginService {
    Database d;

    public LoginService(Database d) {
        this.d = d;
    }
    or
    public void setDB(Database d) {
        this.d = d;
    }
}

```

Main:

```

public class Main {
    Database d = new Postgres MySQL();
    LoginService ls = new LoginService(d);
    SignupService ss = new SS(d);
    UserDetails ud = new UD(d);
}

```

Dependency Injection: Instead of a class creating its dependency itself, it is passed from outside to the class.

Inversion of Control $\Rightarrow$ I was passing the dependency
but IOC is that D.I. is being done by
~~the~~ spring itself.

$\rightarrow$ annotations

@Override

@Component

class ~~login~~ Login Service {

Database d;

login (Database d) {
 this.d = d;

}

}

}